

Another formula in Excel is NPER, which provides a variation on this theme. It works the same way as PMT, but in this case you input how much you wish to spend, and the formula tells you how long your money will last. Moshe Milevsky suggested in his article “It’s Time to Retire Ruin (Probabilities)” from the March/April 2016 issue of the *Financial Analysts Journal* that this formula will be a good starting point for understanding how our spending assumptions impact portfolio sustainability (though he actually expressed the formula in terms of calculus, rather than as an Excel function). NPER is expressed as:

=NPER(rate,pmt,pv,fv,type)

So, reflecting the first problem we investigated above,

=NPER(0.0133,-40109,1000000,0,1)

will give us an answer of thirty years. With a 1.33 percent real return, we can spend an inflation-adjusted \$40,109 from a \$1 million portfolio sustainably for thirty years and be left with nothing at the end of that period.

If we wished to spend \$80,000 and keep the other assumptions, NPER would inform us that our portfolio would only last for 13.6 years before depletion. Milevsky likes to start with this formula because he believes it provides a more useful reality check about a spending plan. Will the portfolio last for as long as you intend? In the context of sustainable spending from available assets, the only way to increase the portfolio’s longevity is to assume a higher investment return or a lower spending amount. Increasing the return assumption by too much endangers the plan, as it will become increasingly difficult for that return to be met in practice.

Incidentally, the RATE function in Excel works in the same basic way as these two formulas, instead telling us the rate of return we would require to meet our **pmt** spending objective for **nper** years. If the implied rate of return is seemingly too high, this formula makes clear that either the level of spending must be reduced or spending will not last for as long as desired.

These formulas provide useful starting points to understand retirement spending because they clearly link spending amounts, rates of return, and